

Math 21 Curriculum

Algebra and its Functions | Middlesex School Mathematics

COURSE AT A GLANCE

Review Content

REVIEW

Primary resources: [Intermediate Algebra 2e 3.2 Slope of a Line](#); [Intermediate Algebra 2e 3.1 Graph Linear Equations in Two Variables](#); [College Algebra 2e 4.1 Linear Functions](#); [Intermediate Algebra 2e 4.1 Solve Systems of Linear Equations with Two Variables](#); [Intermediate Algebra 2e 5.2 Properties of Exponents and Scientific Notation](#)

UNIT LEARNING OBJECTIVES

- I can determine and interpret the slope of a line.
- I can graph a line from an equation and write an equation from given information.
- I can apply a linear function to a context.
- I can identify the solution of a linear system as an intersection point.
- I can solve a simple two-variable linear system using an appropriate method.
- I can simplify an expression using basic positive-integer exponent rules.

Functions and Function Interpretation

REQUIRED

Primary resources: [Intermediate Algebra 2e 3.5 Relations and Functions](#); [College Algebra 2e 3.3 Rates of Change and Behavior of Graphs](#)
Supplemental resources: [Intermediate Algebra 2e 3.6 Graphs of Functions](#); [College Algebra 2e 3.1 Functions and Function Notation](#); [College Algebra 2e 4.1 Linear Functions](#)

UNIT LEARNING OBJECTIVES

- I can evaluate a function from a formula, graph, or table.
- I can determine whether a relation is a function.
- I can determine domain and range from a graph or table.
- I can solve for an input that produces a specified output.
- I can interpret inputs, outputs, domain, and range in context.
- I can calculate and interpret average rate of change.
- I can compare functions represented by formulas, graphs, or tables.

Polynomials and Factoring

REQUIRED

Primary resources: [5.1 Add and Subtract Polynomials](#); [5.3 Multiply Polynomials](#); [6.1 Greatest Common Factor and Factor by Grouping](#); [6.2 Factor Trinomials](#); [6.3 Factor Special Products](#); [6.5 Polynomial Equations](#)

UNIT LEARNING OBJECTIVES

- I can add and subtract polynomials.
- I can multiply polynomials.
- I can factor out a greatest common factor.
- I can factor a quadratic trinomial.
- I can factor a difference of squares and other required special cases.
- I can use factoring to determine the zeros of a polynomial.

Rational Expressions and Equations

REQUIRED

Primary resources: [7.1 Multiply and Divide Rational Expressions](#); [7.2 Add and Subtract Rational Expressions](#); [7.4 Solve Rational Equations](#)

UNIT LEARNING OBJECTIVES

- I can state excluded values for a rational expression.
- I can simplify a rational expression while preserving its restrictions.
- I can multiply and divide rational expressions.
- I can add and subtract rational expressions using a common denominator.
- I can solve a rational equation and reject extraneous solutions.

Exponents and Radicals

REQUIRED

Primary resources: [Intermediate Algebra 2e 8.3 Simplify Rational Exponents](#); [Intermediate Algebra 2e 8.2 Simplify Radical Expressions](#); [Intermediate Algebra 2e 8.6 Solve Radical Equations](#)
Supplemental resources: [College Algebra 2e 2.6 Other Types of Equations](#)

UNIT LEARNING OBJECTIVES

- I can simplify expressions containing negative exponents.
- I can rewrite expressions between radical and rational-exponent forms.
- I can simplify radical expressions.
- I can apply exponent rules to expressions with integer and rational exponents.
- I can solve a basic equation involving a radical or rational exponent and check for extraneous solutions.

Quadratic Functions and Equations

REQUIRED

Primary resources: [9.6 Graph Quadratic Functions Using Properties](#); [9.7 Graph Quadratic Functions Using Transformations](#); [6.5 Polynomial Equations](#); [9.1 Solve Quadratic Equations Using the Square Root Property](#); [9.2 Solve Quadratic Equations by Completing the Square](#); [9.3 Solve Quadratic Equations Using the Quadratic Formula](#); [9.5 Solve Applications of Quadratic Equations](#)
Supplemental resources: [9.6 Graph Quadratic Functions Using Properties](#); [6.5 Polynomial Equations](#)

UNIT LEARNING OBJECTIVES

- I can identify key features of a quadratic function from its graph or equation.
- I can connect standard, factored, and vertex forms to features of a parabola.
- I can solve a quadratic equation by factoring.
- I can solve a quadratic equation using the square-root property.
- I can solve a quadratic equation by completing the square.
- I can solve a quadratic equation using the quadratic formula.
- I can choose an efficient method for solving a quadratic equation.
- I can graph a quadratic function using its vertex, intercepts, and symmetry.
- I can create and interpret a quadratic model for a projectile, area, or optimization problem.

Extension Content

EXTENSION

Primary resources: [8.8 Use the Complex Number System](#); [9.3 Solve Quadratic Equations Using the Quadratic Formula](#)

UNIT LEARNING OBJECTIVES

- I can perform basic arithmetic with complex numbers.
- I can express the nonreal solutions of a quadratic equation using complex numbers.